

Test Specification

NestUp

What is tested, why, and what is not

Internet Technologies — Become a Full-Stack Engineer

RUNI CS 2026 · Final project

Live product nestup-kappa.vercel.app

Repository github.com/nestupweb/nestup

Suite: 97 test files · 740 tests · Vitest + React Testing Library (jsdom) + Playwright (real browser) **Run:** `npm test` — see §9 for the reason that `npx vitest` must never be used directly.

1. What "working" means for this product

Before listing the tests themselves, it is worth stating explicitly what the suite is actually defending, because this determines what is tested and what is not tested.

NestUp can fail in three ways which matter, presented here in descending order of cost:

1. **A member sees the data of another member.** This failure is unrecoverable, because the entire trust proposition of the product collapses.
2. **A core flow silently performs the wrong operation.** For example, a message which does not arrive, a swiped room which returns to the deck, or an edit to a listing which also erases the chats of the owner.
3. **A screen is incorrect or unattractive.** This is embarrassing, but it is inexpensive to correct.

The suite is weighted accordingly. Access control and business rules are tested rigorously. The interface is tested at the level of *behavior*, namely which control appears, what state it is in, where it navigates, and what it renders when the data is empty or missing, rather than at the level of pixels. Those checks are collected in §8.

The following is deliberately not tested: every line of every component, the exact wording of the copy, the exact colors (except in cases where a rule depends on them, as in `map-basemap.test.ts`), and third-party libraries. A suite which asserts on wording eventually becomes a suite which nobody is willing to modify, and a screen which is merely unattractive is caught by looking at it. What the interface must *do* is a different matter, and that is tested.

2. Feature tests — the main flows

Authentication and onboarding

TEST FILE	WHAT IT DEFENDS
<code>auth-form.test.tsx</code>	The sign-up and sign-in forms, including field state, submission and error display
<code>signup-confirmation.test.ts</code>	Registration must not return a usable session. The row is created, a code is mailed, and access waits for confirmation
<code>auth-password-actions.test.ts</code>	The forgot-password and reset flows
<code>auth-confirm-route.test.ts</code>	Every shape of emailed link arrives correctly: signup leads to onboarding, recovery leads to reset, and an invalid link leads to <code>/login</code> with the appropriate notice
<code>auth-email-templates.test.ts</code> , <code>auth-mail.test.ts</code> , <code>mail.test.ts</code>	The mail templates and the sender behave as intended
<code>password-input.test.tsx</code>	The show and hide toggle
<code>redirect.test.ts</code>	The function <code>sanitizeNextPath</code> , which provides open-redirect hardening

Profile

TEST FILE	WHAT IT DEFENDS
<code>profile-action.test.ts</code>	Saving a profile writes the correct rows and invalidates the correct caches
<code>profile-required-fields.test.tsx</code>	Which fields prevent a save, together with the banner which explains this
<code>profile-form-sticky.test.tsx</code>	A rejected form retains whatever the member typed. React 19 resets forms by default, and this test is the guard against that behavior
<code>about.test.tsx</code> , <code>contact-row.test.tsx</code>	Private details and contact visibility
<code>daily-life.test.tsx</code> , <code>daily-life-reminder.test.tsx</code>	The lifestyle questionnaire and the reminder to complete it
<code>profile-amenities.test.ts</code> , <code>profile-safe-room</code> , <code>interests-picker.test.tsx</code>	The preference inputs
<code>profile-avatar.test.tsx</code> , <code>photo-picker.test.tsx</code> , <code>photo-order.test.tsx</code>	Photo upload, ordering and the lightbox
<code>profile-tabs-history.test.tsx</code> , <code>my-listing.test.tsx</code>	The profile tabs
<code>people.test.ts</code>	The public profile of another member

Listings

TEST FILE	WHAT IT DEFENDS
<code>listing-query.test.ts</code>	The browse query, including filters, sorting and pagination
<code>listing-actions.test.tsx</code> , <code>listing-status-action.test.ts</code>	Publishing, pausing and resuming
<code>delete-listing-action.test.ts</code>	Soft removal
<code>mark-taken.test.tsx</code> , <code>listing-taken.test.ts</code>	The "taken" state
<code>listing-title.test.ts</code> , <code>property-tile.test.tsx</code> , <code>listing-gallery.test.tsx</code>	The presentation of a room
<code>filter-bar.test.tsx</code> , <code>sort-menu.test.tsx</code>	The filter and sort controls
<code>save-button.test.tsx</code>	Hearts, both in the signed-in state and in the signed-out state
<code>roommate-tag-picker.test.tsx</code> , <code>roommate-count.test.ts</code> , <code>roommates-fit-rooms.test.ts</code>	The rules governing household composition
<code>co-posters.test.ts</code> , <code>co-posters-action.test.ts</code> , <code>co-poster-invites.test.tsx</code>	Co-ownership and invitations

Matching

TEST FILE	WHAT IT DEFENDS
<code>compatibility.test.ts</code>	The scoring function, including each weighted component and the convention that a missing preference receives approximately 60%
<code>swipe-rank.test.ts</code>	Deck ordering and the <code>MIN_DECK_SCORE</code> threshold
<code>swipe-deck.test.tsx</code>	The deck interface, the like and skip operations, and the empty state
<code>affinity.test.ts</code>	Interest overlap
<code>apartment-prefs.test.ts</code> , <code>no-city-prompt.test.tsx</code>	The requirement of a preferred city, together with its prompt
<code>seed-data.test.ts</code>	Measures the real deck across all 124 cities , so that a town cannot silently become unmatchable

Chat and viewings

TEST FILE	WHAT IT DEFENDS
<code>send-message-action.test.ts</code>	Sending a message, and the idempotent retry of an identical <code>client_id</code>
<code>chat-realtime.test.tsx</code>	The token reaches the socket <i>before</i> the channel joins, and there is exactly one invalidation per burst
<code>chat-visible.test.ts</code>	The per-member "delete chat" cutoff
<code>chat-outbox.test.ts</code> , <code>message-composer.test.tsx</code> , <code>chat-media.test.ts</code> , <code>chat-format.test.ts</code>	Optimistic sending, attachments and formatting
<code>message-owner.test.tsx</code>	The "message the household" entry point
<code>viewing-card.test.tsx</code> , <code>viewing-details.test.tsx</code> , <code>availability.test.ts</code> , <code>calendar.test.ts</code>	Proposing, approving, availability windows and calendar export

Settings and moderation

TEST FILE	WHAT IT DEFENDS
<code>account-actions.test.ts</code> , <code>account-section.test.tsx</code>	Changing the email address and the password
<code>danger-zone.test.tsx</code>	Account deletion, including the listing-handover picker in all three of its shapes , namely no heir, exactly one heir, and several heirs
<code>setting-toggle.test.tsx</code> , <code>settings-gear.test.tsx</code>	The settings controls
<code>moderation.test.ts</code>	Reporting, blocking, unblocking and suspension
<code>notify.test.ts</code>	The new-match notification

3. Invalid-input tests

AREA	WHAT IS ASSERTED
<code>validation.test.ts</code>	Every Zod schema, including required fields, lengths, ranges, enums and coercion
<code>profile-required-fields.test.tsx</code>	A save which is missing a required field is refused together with a field-level message
<code>send-message-action.test.ts</code>	An empty message, an over-length message, a malformed conversation identifier, and a photo path located outside the folder of the conversation itself
<code>listing-query.test.ts</code>	Meaningless query strings fall back to the default values instead of producing an error, because <code>.catch()</code> is applied to every filter
<code>amount-input.test.tsx</code> , <code>phone-field.test.tsx</code> , <code>date-picker.test.tsx</code> , <code>city-combobox.test.tsx</code>	The rejection of non-numeric rent, malformed telephone numbers, impossible dates and unknown cities
<code>photo-rules.test.ts</code> , <code>photo-check.test.ts</code>	An incorrect file type, an oversized file, and a photograph whose content does not correspond to its declared tag
<code>redirect.test.ts</code>	Targets such as <code>?next=https://evil.example</code> , and other off-site destinations, are rejected
<code>image-client.test.ts</code>	The boundaries of client-side compression and of HEIC conversion

4. Business-process tests

These tests assert the rules which make the product itself correct, and not merely the code.

RULE	WHERE IT IS ASSERTED
A swiped room never returns to the deck	<code>swipe-rank.test.ts</code> , <code>cache-invalidation.test.ts</code>
Only rooms scoring 60 or above enter the deck	<code>swipe-rank.test.ts</code>
A seeker who has no preferred city receives no deck	<code>apartment-prefs.test.ts</code>
One person may have one active listing	<code>roommates-fit-rooms.test.ts</code> , <code>co-posters.test.ts</code>
The number of roommates cannot exceed the number of rooms	<code>roommates-fit-rooms.test.ts</code>
Deleting an account transfers a shared listing rather than destroying it	<code>danger-zone.test.tsx</code>
A member who already has a live listing cannot inherit a second one	<code>danger-zone.test.tsx</code>
Deleting a chat hides it for one side only, and the next message restores it	<code>chat-visible.test.ts</code>
A blocked member disappears from the decks and from the search in both directions	<code>moderation.test.ts</code>
A retried message carrying the same <code>client_id</code> is not posted twice	<code>send-message-action.test.ts</code>
Every city is able to fill a deck	<code>seed-data.test.ts</code>

Cache invalidation treated as a business rule

The file `cache-invalidation.test.ts` deserves a separate explanation. It asserts the **scope of impact** of every mutation: that adding a room to the liked list touches exactly `saved:<me>` and `profile:<me>` and nothing further; that no listing mutation reaches into the chat data; and that no chat mutation reaches into the deck or into the room list.

This test exists because of a genuine defect which occurred earlier. Mutations used to respond to every write with a large number of `revalidatePath` calls, and consequently editing a room also discarded the chats and the profile of that member. The test encodes the rule which replaced that behavior.

5. Permission tests

This is the category with the highest value, since the most severe failure mode of the product is the exposure of data across members.

TEST	WHAT IT DEFENDS
<code>cache-invalidation.test.ts</code> , specifically the case <i>"every per-member tag is scoped to the member who acted"</i>	Every <code>deck:</code> , <code>profile:</code> , <code>saved:</code> and <code>chat:</code> tag carries the identifier of the acting member. A tag which omitted that identifier would become a single shared cache key for the entire application , which is precisely the form that a cross-user leak takes
<code>cached-session-boundary.test.ts</code>	No Server Action is permitted to authorize itself using the cached session. Identity is cached for the purpose of <i>rendering</i> only, and every write still performs the uncached check. Both spellings compile successfully, and therefore only a test is able to enforce this boundary
<code>cached-session-boundary.test.ts</code> (continued)	The suspension check still reads the <code>suspensions</code> table, and both gates still redirect on the basis of it
<code>signout-clears-cache.test.ts</code>	Signing out answers with 303 , which forces a full document load, and this is the only mechanism which empties the private caches held by the browser. The test also asserts that there is no GET handler , so that a logout cannot be triggered from another site
<code>member-actions.test.tsx</code>	The Log out button posts to the route handler rather than to a Server Action, since a soft redirect would leave the caches alive
<code>moderation.test.ts</code>	A member cannot remove his or her own suspension, and blocking is mutual
<code>profile-action.test.ts</code>	Only the owner is able to edit a profile
<code>co-posters-action.test.ts</code>	Only a member who was invited is able to accept an invitation

The layer which these tests cannot reach is RLS itself, which is enforced by Postgres, whereas the unit suite mocks the Supabase client. RLS is therefore verified by the real-browser checks described in §7, and by the fact that the policies constitute the *only* path to the data. This is discussed further in the [Security](#) document.

6. Database tests

TEST	WHAT IT DEFENDS
<code>seed-data.test.ts</code>	A sha256 fingerprint computed over the first 92 seed records. Consequently, re-running <code>npm run seed</code> can never silently modify existing demo data
<code>seed-data.test.ts</code> (continued)	Re-measures deck coverage across all 124 cities using the real <code>buildDeck</code> function
<code>city-centres.test.ts</code>	Protects all 124 generated city coordinates against drift
<code>listing-query.test.ts</code>	The translation from filters into SQL, including the arithmetic of the pagination <code>range()</code>
<code>cache-lifetimes.test.ts</code>	Every <code>cacheLife</code> declaration specifies <code>stale ≥ 300s</code> , which is the App Shell threshold, and the router cache is not shorter than the data caches
<code>roommates-fit-rooms.test.ts</code> , <code>roommate-count.test.ts</code>	The denormalized household columns which are maintained by triggers

7. Edge cases

CASE	TEST
An empty deck, either because every room was already swiped or because no room reaches a score of 60	<code>swipe-deck.test.tsx</code> , <code>swipe-rank.test.ts</code>
An empty inbox, and a chat which was deleted and then revived by a new message	<code>chat-visible.test.ts</code>
A room which is removed while a conversation about it is open	<code>listing-taken.test.ts</code> , together with the second <code>listings</code> <code>SELECT</code> policy
Concurrent creation of a conversation, producing a unique-constraint race	<code>findOrCreateConversation</code> falls back to re-reading the row
A duplicated message send	<code>send-message-action.test.ts</code>
Account deletion with zero, one or several eligible heirs	<code>danger-zone.test.tsx</code>
The realtime socket joining before the token arrives	<code>chat-realtime.test.tsx</code>
A failing realtime synchronization must not throw inside an effect	<code>chat-realtime.test.tsx</code>
A stale <code>?needs=cities</code> parameter remaining in the URL after the member has already corrected it	<code>no-city-prompt.test.tsx</code>
A listing which has no photographs	<code>listing-gallery.test.tsx</code>
Navigation must remain within a single tab	<code>same-tab-navigation.test.ts</code> together with <code>npm run check:nav</code>

Real-browser checks (Playwright)

Certain properties cannot be asserted inside jsdom, because they require a real engine, a real GPU canvas, or the real deployment.

SCRIPT	WHAT IT PROVES
<code>npm run check:map</code>	Both themes render correctly, and the script counts red pixels directly from the WebGL canvas , since a GL layer has no DOM which could be queried
<code>npm run check:nav</code>	No internal link opens a new tab
<code>.superpowers/e2e/nav-cache.mjs</code>	21 assertions executed against the live site: signing out destroys the JavaScript context; <code>/swipe</code> redirects after a logout; the back button cannot restore a signed-in page; and a second member using the same tab sees only his or her own data
<code>.superpowers/e2e/chat-freshness.mjs</code>	A real message reaches the thread <i>and</i> updates the cached inbox row, and it survives navigating away and returning
<code>.superpowers/e2e/skeleton-duration.mjs</code>	Measures how long a loading skeleton is actually visible on screen, per tab

8. Basic UI tests

The interface is tested for what it does rather than for how it looks. Each of these runs in React Testing Library against real user events — clicks, typing, arrow keys, Escape — and asserts on what a member would actually see or reach.

Navigation chrome

TEST FILE	WHAT IT DEFENDS
<code>bottom-nav.test.tsx</code>	The four destinations render, the current one is marked, nested routes keep their tab active, the unread badge appears inside the Chat link and nowhere else, and the bar hides itself on a small screen inside an open thread and on the auth pages
<code>back-button.test.tsx</code>	The button names the page it will return to, pops the in-app trail instead of growing it, survives a reload through <code>sessionStorage</code> , and falls back to the list page when there is no history to return to
<code>nav-direction.test.ts</code>	Moving right along the tab bar is forward and moving left is back, which is what drives the transition animation
<code>same-tab-navigation.test.ts</code>	Every internal link stays in the same tab, which is also enforced in a real browser by <code>npm run check:nav</code> (§7)
<code>member-actions.test.tsx</code>	Log out posts to the signout route rather than to a Server Action, and the settings link sits beside it

Form controls

TEST FILE	WHAT IT DEFENDS
<code>date-picker.test.tsx</code>	Typing <code>dd/mm/yyyy</code> inserts its own slashes and caps each field, impossible dates are refused, the calendar blocks days before <code>min</code> and outside the allowed weekdays, and typing cannot bypass a day which the calendar itself blocks
<code>amount-input.test.tsx</code>	Thousands are grouped while typing but submitted bare, an empty field submits nothing rather than a zero, and the caret stays put when a comma appears to its left
<code>phone-field.test.tsx</code>	A country is searched the way a city is, namely type, arrow and Enter, the flag and dial code follow, a stored international number pre-selects its own country, and Escape closes the list
<code>city-combobox.test.tsx</code>	The chevron opens every city A–Z grouped by letter, typing filters to <i>every</i> match rather than to the first eight, and Escape closes the list
<code>sticky-form.test.tsx</code> , <code>profile-form-sticky.test.tsx</code>	A rejected form keeps every value the member typed. React 19 resets forms by default, and this is therefore the guard against losing a long profile to one validation error
<code>password-input.test.tsx</code>	The show and hide toggle

Theme and appearance

TEST FILE	WHAT IT DEFENDS
<code>theme-toggle.test.tsx</code>	The control is a real switch, it flips the dark theme on <code><html></code> , it persists to <code>localStorage</code> , and it reports the correct state when the page loads already in dark mode
<code>map-basemap.test.ts</code>	Both themes resolve to the correct basemap, which is verified against a real GPU canvas by <code>npm run check:map</code> (§7)

Empty, missing and degraded states

TEST FILE	WHAT IT DEFENDS
<code>listing-gallery.test.tsx</code>	A room which has no photographs still renders
<code>swipe-deck.test.tsx</code>	An exhausted deck shows its empty state rather than a blank screen
<code>listing-match-score.test.tsx</code>	A Listings row shows the same two scores as the swipe deck, a social score of <code>null</code> degrades to an em dash rather than to a zero, a signed-out visitor sees no score pills at all, and a low score is shown rather than hidden
<code>property-tile.test.tsx</code>	A plain tile is merely a link, whereas a liked tile carries a filled heart which toggles on tap without following the link underneath it
<code>room-map-button.test.tsx</code> , <code>map-explorer.test.tsx</code>	The map opens on demand rather than on page load, the legend names only the categories actually present, a failed lookup offers a retry instead of an empty map, and closing is possible by button, by Escape and by backdrop, with focus returning to the trigger
<code>.superpowers/e2e/skeleton-duration.mjs</code>	Measures, in a real browser, how long each loading skeleton is actually visible on screen (§7)

9. How to run the suite

```
npm test                # 97 files, 740 tests
npm test -- --testTimeout=25000  # use this if userEvent tests time out (see below)
npx tsc --noEmit        # types
npx eslint              # lint
```

There are two environment traps, both of which were discovered through experience, and both of which are worth knowing before accepting a red result as genuine:

1. **Always use `npm test`, and never `npx vitest run`.** The npm script sets `NODE_OPTIONS=--no-experimental-webstorage`. Without this flag, Node 25 installs its own inert `localStorage` global which jsdom does not replace, and consequently `theme-toggle.test.tsx` fails with `localStorage.clear is not a function`, which is a failure that says nothing at all about the component itself.
2. **OneDrive synchronization starves `userEvent`.** The project resides in a synchronized folder, and when OneDrive is busy, approximately half a dozen interaction tests exceed the default limit of 5 seconds. The suite should therefore be re-run with `--testTimeout=25000` before a timeout is treated as a real failure.

10. Known gaps

These are stated plainly, because a test document which claims complete coverage is not credible.

- **The RLS policies are not exercised by the unit suite.** The suite mocks the Supabase client, and therefore the policies themselves are covered only by the real-browser checks and by manual review. A future improvement would be a test which executes real queries as two different members against a dedicated test project.
 - **There is no automated end-to-end journey.** Playwright is used for targeted checks rather than for a complete script covering registration, profile creation, swiping, chatting and scheduling a viewing.
 - **Email delivery is mocked.** The templates and the sender are tested, whereas actual arrival in an inbox is verified manually.
 - **The Gemini photo check is mocked.** Its contract is tested, but the judgment of the model itself is not.
 - **Load has not been measured.** The reasoning about scale is analytical; see the [Scale](#) document.
-